NeuralRetail is an enterprise-grade AI Sales Intelligence Platform designed to transform raw retail transaction data into actionable business insights. It leverages a dataset of over 1 million transactions to provide:

* Demand Forecasting: Predicting future sales to optimize inventory.
* Churn Prediction: Identifying customers at risk of leaving.
* Customer Segmentation: Grouping customers based on buying behavior (RFM).
* Revenue Intelligence: Real-time analytics and KPI tracking.

How It Works: The 6-Step Pipeline

The system is orchestrated by a central runner (run.py) that executes the following sequence:

Step 1: Data Ingestion & Cleaning (ingest.py)

* Goal: Move data from the Raw layer to the Silver layer.
* Process:
* Loads raw CSVs (transactions.csv, customers.csv, etc.).
* Cleans data by handling missing values, removing duplicates, and correcting data types (e.g., date parsing).
* Produces transactions_clean.csv, which serves as the "single source of truth."

Step 2: Feature Engineering (features.py)

* Goal: Transform cleaned data into mathematical features for ML models.

* Process:

* RFM Analysis: Calculates Recency, Frequency, and Monetary scores for every customer.

* Temporal Aggregation: Creates daily sales aggregates for forecasting.

* Behavioral Features: Calculates average basket size, weekend shopping ratios, and tenure.

Step 3: Churn Prediction Model (churn.py)

* Model: XGBoost Classifier.

* Logic: It uses a temporal split (Oct 2011 cutoff). It looks at customer behavior before the cutoff to predict if they will make a purchase in the following 70 days.

* Key Features: Days since last purchase, total spend, and number of unique products bought.

Step 4: Demand Forecasting (forecast.py)

* Model: An Ensemble of Gradient Boosting (GBM), Ridge Regression, and Prophet.

* Logic: It treats sales as a time-series problem. It uses "Lags" (what happened 7 and 30 days ago) and "Rolling Windows" (the average of the last week) to predict future demand.

* Accuracy: Achieves a MAPE (Mean Absolute Percentage Error) of ~35%, which is strong for retail data containing volatile holiday spikes.

Step 5: Customer Segmentation (segmentation.py)

* Model: K-Means Clustering (with PCA for dimensionality reduction).
* Logic: Groups customers into 3-5 distinct segments (e.g., "Champions," "At-Risk," "Potential Loyalists") based on their RFM scores.

Step 6: Visualization & Export (powerbi_export.py)

* Goal: Make the data accessible to business users.
* Outputs:
* Power BI: Exports processed datasets to the powerbi_data/ folder.
* Reports: Static charts (ROC curves, feature importance) saved in reports/.

Technical Stack

* Logic: Python (Pandas, Scikit-Learn, XGBoost, Prophet).
* API: FastAPI (15+ endpoints for real-time predictions).
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Project Structure

text

NeuralRetail/

??? data/   # Raw, Silver, and Gold data layers

??? src/   # Core pipeline scripts (ingest, features, models)

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??? models/     # Saved .pkl model files
??? run.py      # Master execution script
??? api/        # FastAPI implementation
```